



Basic Details of the Team and Problem Statement

Ministry/Organization Name/Student Innovation: Ministry of Jal Shakti

PS Code: 1293

Problem Statement Title: Automatic regulation of valves for release of water based upon soil moisture availability in the root zone of the crop, using artificial intelligence, in a piped and micro irrigation network of irrigation system.

Team Name: SMARTIrriS AI

Team Leader Name: Himanshu Mangilal Soni

Institute Code (AISHE): C – 41227

Institute Name: AISSMS College of Engineering, Pune

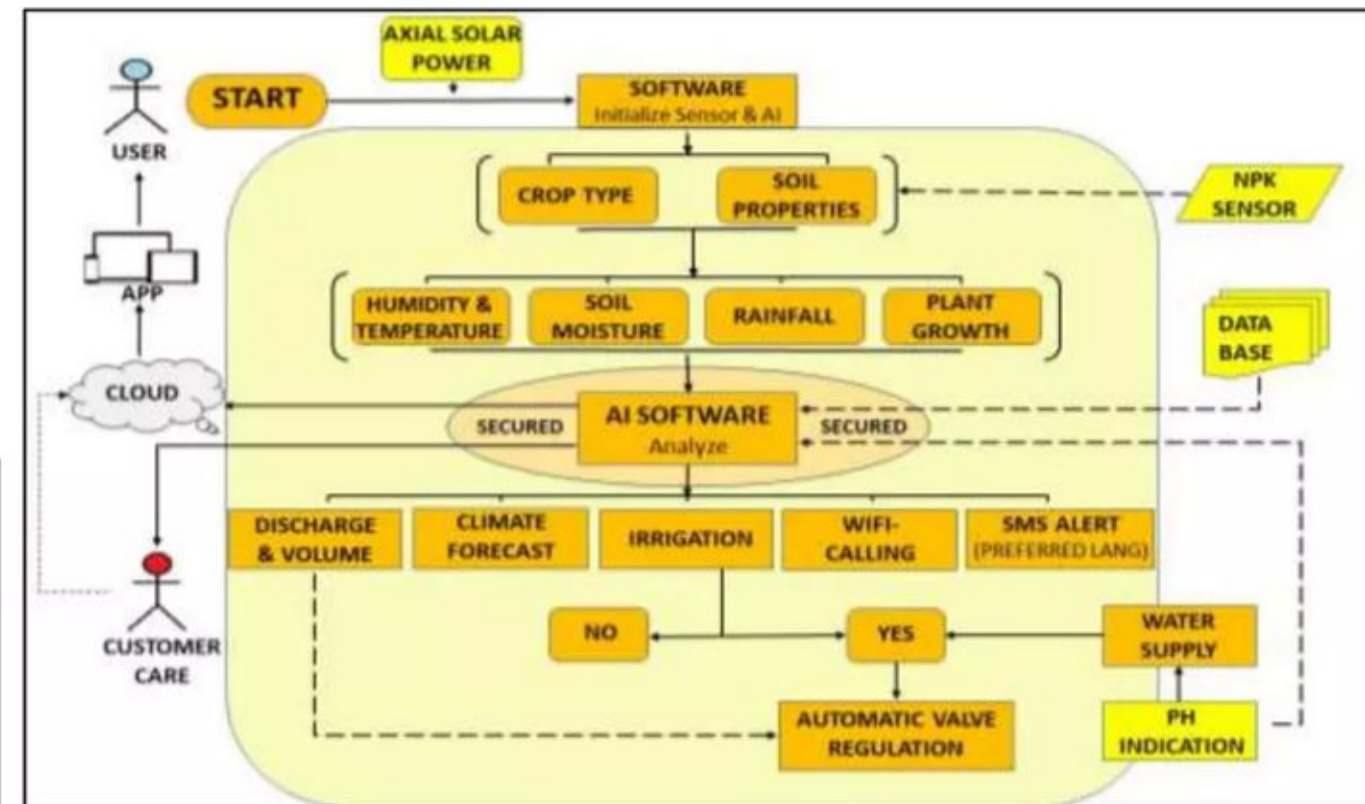
Theme Name: Hardware

Idea/Approach Details

Describe your idea/Solution/Prototype here:

The prototype is going to automatically regulate the valves based on the soil moisture content in the root zone of the crop by developing AI Software

- Interface will be easy for the farmers
- Automatic opening of valve based on discharge & volume
- Decision of supply of water to be taken by AI tool based on dataset
- Fetching Data of Moisture, Rainfall, Growth, Temperature & Humidity
- Soil properties & crop type informed using NPK sensor
- Ph indication of water to be given
- Solar Power source for operation of sensor and centralized system w.r.t. sun rays direction



Describe your Technology stack here:

- Cloud to be used to store and access data – AWS
- Raspberry Pi 4 / Orange Pi 5
- MS Windows, python, C++
- Valve Regulation – AI / ML supervise Learning (ANN)
- Using python libraries – Numpy, Panda, Tensorflow
- Data base formation in CSV Format.

Idea/Approach Details

Describe your Use Cases here

- Realtime Data
- Previous Data Stored
- SMS Alert & Dashboard Notification
- Secured Software
- Emergency Calling / Wi-Fi Calling
- Customer Care Support
- Weather Forecast
- Pump & Valve - on / off indication
- Pump & Valve controls - Automatic & Manual
- Water PH indication

Describe your Dependencies / Show stopper here

- Sensor – Moisture, NPK
- Wi-Fi
- Solenoid Valve
- Dataset of Soil Properties, Moisture, Plant Growth and Windy App
- Power Backup

Team Member Details

Team Leader Name: Himanshu Soni (B.E.)

Civil Engineering

III

Team Member 1 Name: Aryan Tilekar (B.E.)

Civil Engineering

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Team Member 2 Name: Prachi Tonde (B.E.)

Civil Engineering

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Team Member 3 Name: Sounak Ray (B.E.)

Computer Science Engineering

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Team Member 4 Name: Yuvraj Kottalagi (B.E.)

Computer Science Engineering

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Team Member 5 Name: Ashish Wanve (B.E.)

Electronics & Telecommunication Engineering

III

Team Mentor 1 Name: Dr. Vidya N. Patil

Category: Academic

Expertise: Innovation / Automation

Domain Experience: 23 Years

Team Mentor 2 Name: Dr. Atulkumar A. Manchalwar

Category: Academic

Expertise: AI, Python Programming in Civil Engg.

Domain Experience: 10 Years

Important Pointers

Please ensure below pointers are met while

- Kindly keep the maximum slides limit to 4 pages
- All the topics should be utilized for description of your idea
- Try to avoid paragraphs and post your idea in points
- Keep your explanation precisely and easy to understand
- Idea should be unique and novel. If it has a business potential more weightage will be given.
- Apart from this PPT abstract of your idea will be asked separately while submitting
- You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported
- You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.